

S.P.A.C.E. – Studying perceptions, activities, connections in extreme environment

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ABSTRACT

Extreme environments place high demands on the mental well-being of their inhabitants. As a case study, this interdisciplinary research uses the *LunAres Research Station* to examine the relationship between the built environment, perception and behaviour of its inhabitants, and their overall well-being. The study demonstrates that the emotional and aesthetic dimensions of lived-in spaces significantly influence daily activities and psychological well-being. We thereby extend research from everyday domestic and urban/social spaces to extreme environments such as extra-terrestrial habitats, where previous research has traditionally prioritized functionality and life support.

The study is based on data collected during seven simulation missions at the LunAres Research Station in Poland between 2023 and 2024. We employed an interdisciplinary approach integrating architecture, philosophy, and psychology to develop a comprehensive understanding of perceived habitability.

We investigated (1) how individuals emotionally and aesthetically relate to a simulated astronaut environment, and (2) how these relations connected to daily activities, including social interactions and work-related functioning, as well as overall well-being. We applied a mixed-method approach to collect quantitative and qualitative data; combining research techniques drawn from our respective fields. Moreover, by employing a daily diary technique we tracked the participants appraisals and subjective assessment of activities and well-being in relation to the habitat throughout the two-week missions.

The results suggest that while 'liveability' and functionality ratings correlated with self-reported well-being assessments as well as overall work or social activity success, the emotional attachment to the space was equally, and in some cases even more, correlated to well-being. These findings highlight the importance of considering emotional and aesthetic qualities alongside functional requirements in the design of habitats for isolated and confined environments. Beyond their relevance for future space habitats, they also contribute to a broader understanding of how the built environment shapes human experience and provide insights for creating more supportive, inclusive, and psychologically sustainable environments on Earth.

1. Introduction

The relationship between the built environment and human psychological well-being has received increasing attention across architecture, environmental psychology, and human factors research. This is equally relevant in isolated and confined environments (ICEs), where the design of habitats directly influences crew performance, health, and mission outcomes. Previous studies have investigated the influence of

the built environment on psychological well-being, emotional responses, cognitive performance, and human behaviour in isolated and confined environments (ICEs) and space habitat analogues (e.g., Ref. [1–5]).

In the presented interdisciplinary study, we explored the influence of built environments on human activities, with a particular focus on the emotional dimension of individuals' perceptions of and engagement with their immediate surroundings. With this research project, we seek to understand how individuals perceive the physical spaces they inhabit and how these perceptions manifest at the level of feelings, emotional

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Acronyms/Abbreviations

Analog Habitat for Lunar or Martian missions in Poland (LunAres)
 Ecological Momentary Assessment (EMA)
 International Space Station (ISS)
 Extra-terrestrial Activity (EVA)
 Isolated and Confined Environments (ICE)

attachments, or other phenomenal responses. The working argument for this project is that emotional and aesthetic aspects—perhaps even above and beyond basic infrastructural requirements for support of work and life—may be a key support or hindrance to both individuals' daily activities and to their well-being. Thus, knowledge in this area will support more efficient evaluation and planning of extreme inhabited environments, crew preparations, and mission success.

This paper presents the interdisciplinary study approach and its rationale, the study method and preliminary findings, with a focus of the results from daily diary survey data.

2. Rationale for interdisciplinary research

This research embraced an interdisciplinary approach, uniting experts from diverse fields, including Architecture and Design in Extreme Environments, Psychology, Experimental Philosophy, Neurourbanism, Habitability, and Emotional connections to place. This collaborative effort was made with the goal to provide comprehensive insights into the intricate and multifaceted nature of perceived habitability in isolated and densely populated environments. The study encompassed three key research areas:

Data Collection: Firstly, the study collected and analysed individual emotional aspects of spatial perception, to understand how individuals emotionally engage with the built environment, how this may impact daily human activities, as well as which particular aspects of their living space may drive these responses.

Practical Consequences: Secondly, the study investigated the practical consequences of emotional space perception. By analysing the relationship between perceived space and daily human activities (sleep, hygiene, food, work and leisure; cf [2]) the study intends to identify patterns and correlations of human activities that can inform architecture and also urban planning practises to create environments that enhance human activities.

Variations in Space Perception: Third, the study recognized the importance of inclusivity and equity in our built environments. We will also highlight potential variations in space perception.

Beyond addressing fundamental infrastructure needs, our research posits that the emotional dimension plays a central role in day-to-day well-being and resilience, significantly impacting various aspects of work and interaction. This investigation becomes especially crucial in extreme environments where the design of habitable spaces demands utmost care and innovation.

3. Study approach

Historically, the primary focus of space habitats design has been on functionality and the efficiency of available space, infrastructure, and equipment. The reason for this emphasis is abundantly clear. Due to constraints such as transportation, extreme environmental conditions often immediately beyond the parameters of a built environment, and an overall emphasis on a specific mission or functional purpose for the design itself, extra-terrestrial habitats are rightfully compact and space-limited, making re-configuration or adaption challenging once launched. Therefore, space habitats are always the result of an

optimization process - a trade-off between mission-related constraints and the goal of sustainable missions and architecture design. However, the '*Risk of Incompatible Vehicle/Habitat Design*' has been identified by the NASA Human Research Program (HRP) as a recognized risk to human health and performance in space [6]. Since future space habitats will inevitably remain constrained in terms of volume, mass, and opportunities for modification, a deeper understanding of the reciprocal relationship between people and the built environment—and of how space is perceived and experienced—is essential for designing habitats that support both operational performance and long-term well-being.

Recommendations for habitat design have been made by professionals across disciplines, including engineers, architects, designers, mission planners, psychologists, astronauts, and scientists.

They address a broad range of habitability aspects, including spatial organization and interior layout, workflow and operational efficiency, privacy and territoriality, lighting, colour, design and placement of windows and views, crew autonomy, and opportunities for personalization, all of which have been shown to influence health, well-being, behaviour, and mission performance [2,3,5–14].

Previous research has further examined the relationship between habitat design, crew activities, and perceived habitability, demonstrating that built environments influence not only operational performance but also socio-spatial behaviour, emotional responses, cognitive functioning, and psychological well-being [3–5,15] [16]. Still, some gaps remain in understanding these relationships over extended time. Our research focuses on the following areas:

1. Emotional experience of the built environment

Comparatively little is known about how emotional attachment, aesthetic appreciation, and subjective perceptions of the built environment influence perceived habitability and psychological well-being in isolated and confined environments.

2. Psychological adaptation to challenging environments

Extreme, confined, isolated and dense environments pose unique challenges to individuals' psychological well-being. A better understanding is needed of how individuals perceive and emotionally adapt to the built environment, and how these perceptions influence coping strategies, everyday activities, and overall well-being over time.

3. Relationship between habitat design and everyday activities

Although habitat design recommendations exist, there is limited empirical evidence linking specific spatial qualities with everyday activities, social interactions, work performance, and perceived habitability over time.

4. Socio-spatial dynamics in isolated and compact spaces

Confined environments require continuous negotiation of privacy, territoriality, and shared space. More research is needed to understand the reciprocal relationship between the physical environment and social interactions, including how habitat design influences crew cohesion, community building, and social isolation.

5. Innovative and evidence-based design solutions

The study may reveal emotional responses that contribute to further concrete design solutions. There's a need to bridge the gap between emotional understanding and practical architectural and urban planning recommendations that can enhance the habitability of extreme and dense spaces.

While the study acknowledges the importance of inclusivity, more specific gaps exist concerning the design and accessibility of built environments for individuals with disabilities. Other knowledge gaps that this project may also inform include: cross-cultural comparisons of responses; resilience strategies of people to challenging environments; environmental sustainability strategies based on emotional responses to behaviours; mental health implications in extreme environments over time; temporal variability of emotional response in built environments, and interaction with technology in confined environments.

3.1. Study innovation

On a broader level, this project aims to contribute to the ongoing discourse regarding our engagement with the built environment, both in outer space and on Earth. There are significant differences between these two contexts. Most environmental, physical, psychological, and organisational aspects are not directly comparable. However, when examined from a human-centered perspective, intriguing similarities emerge (cf. [3]).

The principal function of any human habitat is to provide a living and working environment for humans. Thus, generally-speaking human requirements for sustaining life, work, and well-being remain relatively constant across different environments. Regardless of the environment, a human must sleep, use the toilet, eat, socially interact, and be active in some way (see also [2,17]).

What is also central yet often overlooked are people's affective-aesthetic relations to the environment, such as a sense of belonging, identity and emotional connection [18] often related to *finding an environment interesting and beautiful* [19]. Once again this suggests that, beyond mere “functionality” (i.e., addressing the basic infrastructure needs of inhabitants), the affective-aesthetic dimension may be a key driver of day-to-day well-being and resilience. It may also have specific impacts on many work-related and interactional factors.

In discussions of everyday life architecture and especially urban planning, the topic of individuals' aesthetic and emotional relations to a space has been a growing area of emphasis as researchers and planners have looked beyond traditional infrastructural or “liveability” metrics to other potential drivers of individual and society well-being and flourishing [20–23]. Especially this has involved renewed emphasis on psycho-spatial factors involving how people perceive and relate to their spaces. Among these, researchers have highlighted features such as appraised sense of beauty, interestingness meaningfulness, or clarity of organization (e.g., Ref. [23,24]), all of which are argued to relate to preference and especially day-to-day positive mood and functioning [25, 26]. Also especially notable, emerging studies have suggested the emotional dimension, or how we feel and relate to a space, may be a key driver of day-to-day well-being and resilience. This is mentioned in relation to topics such as “placemaking” [23,27], prioritization of “minds and souls” in places and experiences [28,29] or a feeling of attachment or sentiment inspired by city spaces (e.g., Ref. [30,31]). This contains both an element of contentment, comfort, and cosiness as well as an element of variety and fascination [23] that have also been shown in empirical studies to relate to well-being features [32], see also e.g., Ref. [33]). It may also have specific impacts on many work-related and interactional factors.

These findings are particularly relevant for space habitat design. Unlike terrestrial buildings, the importance of habitability design is of even greater importance. Space habitats are characterized by severe constraints in terms of mass, volume, transportation, and opportunities for modification once deployed. Within compact and limited spaces, the design and architectural interventions needs ultimate care. Every design decision must fulfil multiple functional, social, and psychological requirements. A better understanding of how inhabitants emotionally perceive and interact with their environment therefore provides an additional design dimension that can improve habitability without fundamentally changing the physical constraints of the habitat.

4. Setting

The research was carried out in the frame of selected LunAres Missions between 2023 and 2024. The LunAres Research Facility (Fig. 1), in the city of Pila, Poland, was established in 2017. It is a specialized research facility for conducting simulated human space missions, referred to as space mission analogue studies.

The facility provides full isolation, allowing for complex research on the psychological and physiological impact of long-term extra-terrestrial



Fig. 1. View of the LunAres Habitat entrance (Credit: LunAres).

human presence. The general objective of LunAres is to create a research platform to support scientific and technological development in human space exploration [34]. Additionally, the character of LunAres Research Station - being an analogue habitat - provides researchers with a great opportunity to combine various scientific studies (e.g., medical and psychological) with the space mission scenarios.

The LunAres station offers 176 m² of habitable space, designed to accommodate crews of up to six people. The station consists of seven interconnected modules, all surrounding a central domed atrium covering approximately 300 m² (Fig. 3). Each module serves a distinct function, including dormitory, kitchen, gym, storage, laboratory, workshop, sanitary module, office, and the atrium, which is equipped with an airlock (Fig. 2).

5. Methods

The study employed a mixed-method approach and collected quantitative and qualitative data, combining various research techniques drawn from the respective fields of the research team, including architecture, philosophy, and psychology. For the present report, we focus primarily on the daily diary data. The quantitative survey data formed the primary basis for the statistical analysis, the qualitative data from the post-mission interviews were used to contextualize and interpret the quantitative findings.

5.1. Participants

The investigation was conducted with a total of 33 participants during seven missions in 2023 and 2024 (ICAres-2-M01-3). Each mission lasted 14 days, with crew sizes ranging from 4 to 8 members (see Table 1). The crews comprised of both females and males, representing diverse nationalities from across Europe, America, and Africa.

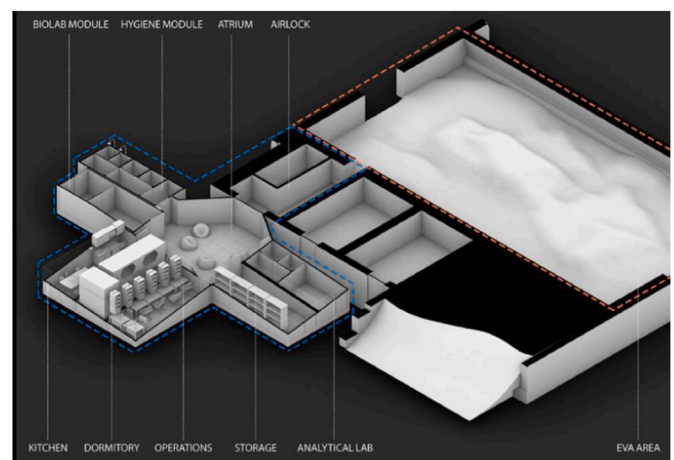


Fig. 2. Overview of the habitat layout. (Author: Matjaz Tancic).



Fig. 3. The atrium - multifunctional area where many crewmembers spend most of the days as “it was the most spacious, liveable space where we could all have enough space for ourselves” (Foto by Sara Borjanovic).

Table 1
Overview of participant sizes in specific missions and corresponding dates.

Mission ^a	date of mission	participants
1	Oct.23	6
2	Nov.23	8
3	Feb.24	5
4	Mar.24	5
6	May.24	5
7	Jun.24	4
Total		33

^a Note, mission 5 omitted due to issue with data collection via online survey form.

Demographic data were not systematically collected for the purposes of this study. However, mission records indicate that participants ranged in age from approximately 21 to 58 years. All participants completed the study as part of their required daily tasks related to their research mission.

5.2. Procedure and study design

A mixed-method approach was used to gather both qualitative and quantitative data. The questionnaires and methods were developed through multiple iterations by the entire research team, aiming to complement previous studies and integrate the team's multi-disciplinary perspectives. This included the following at the specified time points.

(1) **Behavioural questionnaires** were administered at three time points (pre-mission (specific date varying between participants), Day 2, midpoint (day 8), end (days 12-13). These were used to assess expectations and self-reports of certain issues. While this data evaluation is ongoing, and not fully discussed in this paper, we will provide some preliminary examples particularly related to baseline expectations. Participants completed the questionnaires online at predefined time points during the simulation as part of the mission schedule. All were administered via an online survey using the commercial software programs SurveyMonkey and Qualtrics (Qualtrics LLC, Provo, USA) and included the following items:

Scale-based and short answer questions to assess expectations for mission activities and specific factors (pre-mission): *What are you looking forward to?* (Open-ended); *What do you expect to be the most challenging part of this mission for you, on a personal level, and for the whole group?*

(Open-ended); *Please shortly describe the habitat. What do you like best? What do you like least?* (Open-ended); *Do you expect to have difficulties with any of the following activities?* Please rate from 1 to 7. If yes, please describe why (items = Sleep, Hygiene, Food, Work, Leisure).

In addition, participants also answered regarding a number of well-being and quality of life factors (all time points). These were based on standardized batteries but shortened by our team to represent only single items judged to be most representative of the target question (following [23]). All questions used the same format, “Rate your current sense of [X] on a scale of 0-7 (0 being lowest, 7 being highest),” for the following items: current sense of wellbeing, quality of life, ability to regulate your emotions, social connectedness, loneliness.

Participants also completed assessments of their expected emotional attachment (pre): *To what extent do you expect to have an emotional attachment to the habitat on a scale of 1-5* (1-very negative, 5 very positive. (Rating).

For the Day 2, 8, and end surveys, questions were repeated in addition to assessments of work and life aspects: *Please describe your daily work routine. What were your main work tasks? Where did you work most of the day? Did you work together with other crewmembers? Which issues come to your mind?* (Open-ended); *Please describe your daily after-work routine. What were you doing when you didn't work? Which activities were your favourites and with whom and where did those take place?* (Open-ended); *Which activities do you enjoy most? With whom and where do these activities take place?* (Multiple Choice, with comments: Sleep, Hygiene, Food, Work, Leisure).

(2) Some participants also engaged in **semi-structured post-mission interviews** between 30 and 60 min to provide a more nuanced assessment of their experiences and assessment of the space. In the interviews participants' personal experiences of the habitat and their emotional reactions to it were investigated. Discussion themes include their thoughts on spatial attributes, their daily habits, social engagements, and factors that – they found - facilitated or obstructed well-being during the mission. Interviews were performed with an open-ended interview framework, enabling participants and also the interviewer to expand on matters they deemed especially interesting. The interviews enhanced the quantitative data by offering extended insights into the lived experiences inside the habitat and the significance participants assigned to various spatial attributes.

The interview themes included background and mission experience, human activities, habitat design and architectural perception, crew dynamics and social interaction, diversity and inclusion, and well-being and recommendations for future habitat design. Example questions included: *How did you experience the habitat during the mission? Which spaces did you enjoy most or least, and why? Were there architectural features that supported or hindered your daily activities? How did the habitat influence social interactions, privacy, or conflict resolution? What design improvements would you recommend for future space habitats?*

(3) Finally, participants were asked to complete a **Daily Diary** during the first 12 days of their mission in which they reported on their perceptions, and feelings specifically that day, which provided the main assessment for the present paper.

The Daily Diary represents one example of a Daily life method, often placed under the umbrella term “Ecological Momentary Assessments” (EMA), and which represent an alternative approach to traditional pre/post surveys [35,36]. By asking individuals to make periodic reports, they allow for more longitudinal and ecologically valid assessments, tracking opinions, feelings, experiences, and behaviours of individuals throughout their everyday lives. Repeated measures of the same questions allow for better modelling of person- or context-related variations and have been shown to be particularly effective in tracking the effects

of interventions, including environmental engagement, especially in terms of aesthetic aspects (e.g., see Ref. [35]) or a recent paper by members of our team, [37].

In our study, the daily diary method was implemented using an online survey tool, scheduled to be completed each day at 7 p.m. The survey took approximately 5–10 min to complete and included 20 questions, asking participants to reflect on their thoughts and feelings that day.

In Questions 1–7, participants were asked to report on several aspects of their general wellbeing in the environment, using the prompt: “Today, how were your feelings in relation to ... ?” (1 = very low, 5 = very high). Topics included Well-being, Quality of Life, Social Connectedness, Emotion Regulation, Loneliness, Social Interaction, and Anxiety.

In Questions 8–12, participants were asked to report on their daily life activities relating especially to their actions as a member of the LunAres team: “Today, how were your feelings in relation to ... ?” with the same scale (1 = very low, not satisfying; 5 = very high, satisfying). In this case, we referred to activities of Sleep, Hygiene, Food, Work, and Leisure.

Questions 13–18 focused on psycho-spatial assessment, where participants rated their emotional connection to the space for that day. This question asked: “Today, how strong was your emotional attachment to the space?” (0–100, with 100 being very high). Additionally, participants were asked to assess other aspects of the space, including livability (“How livable was the space?”), functionality, and hedonic qualities such as beauty, interest, and lovability.

Lastly, Questions 19–20, participants were asked to identify which activities posed the most difficulties and which were the most pleasant.

5.3. Collected data and assessment

The data collected comprised both qualitative interview segments and scale-based or short-answer responses. For the present report, we focus primarily on the daily diary data. This was assessed via a combination of descriptive statistical analyses as well as, to measure the main relationships between space appraisals and well-being/daily functioning outcomes, use of inferential statistics. This is supplemented by some quotes from the interviews that support findings. Statistics were assessed using, for basic descriptives and boxplots, JASP, and for correlations, the R package. R Core Team. (2023). R: A Language and Environment for Statistical Computing. R Foundation f.

6. Results

All participants completed a sufficient number of daily reports (i.e., with missing data at the participant level of maximum 16%; 2/12 days). Thus, all were retained for analysis (see Ref. [38]), with missing days omitted at the cell-wise level.

6.1. Participants and pre-mission expectations

A selection of participants' expectations regarding the quality of work, well-being, and possible emotional attachments is shown in Figs. 4–7. These data are taken from a preliminary assessment of the pre-mission questionnaire responses (n = 19).

On average, participants expected low to moderate difficulty in performing their work (Fig. 4). However, they also showed somewhat low expectations for well-being (Fig. 5), presumably due to some nervousness about the upcoming missions. Participants expressed neutral expectations regarding their emotional connection to the space (i.e., neither positive or negative) (Fig. 6). Interestingly, despite these neutral expectations regarding emotional valence, participants showed rather strong expectations that they would develop an emotional attachment to the space, even before entering the missions (Fig. 7).

Also, of note, we observed a wide range of responses among individuals, with participants covering most points on the given scales.

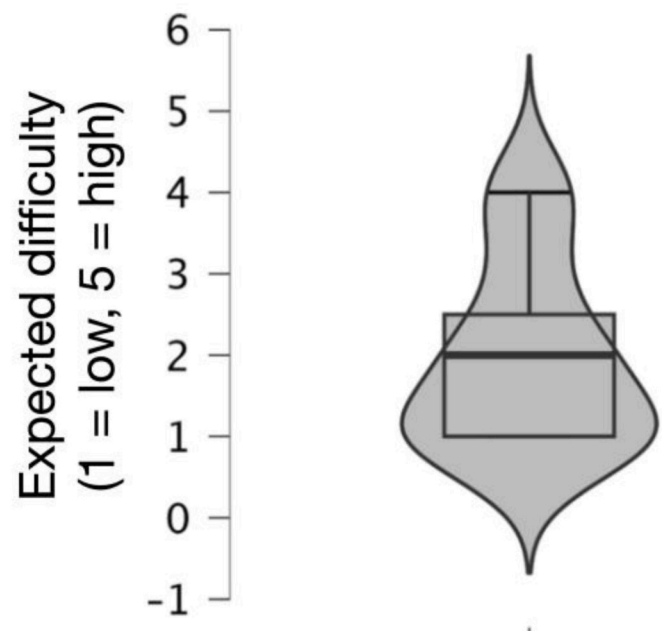


Fig. 4. Participants' expectations regarding the Quality of Work. (N = 19).

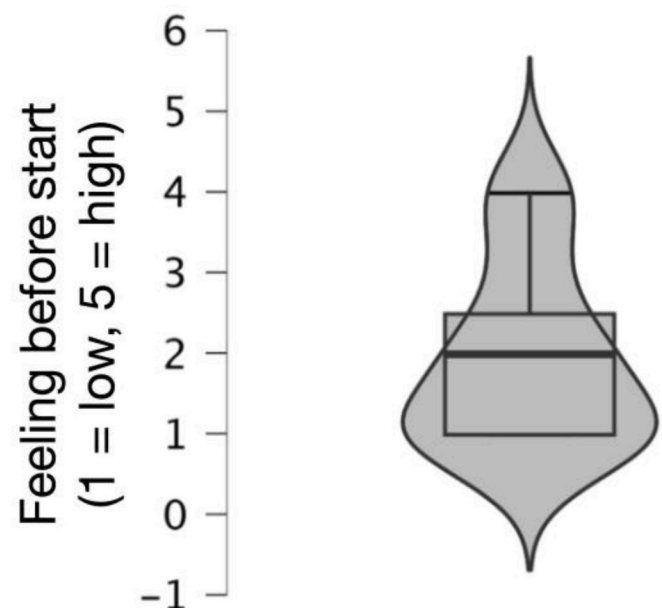


Fig. 5. Participants' expectations regarding their Well-being. (N = 19).

6.2. General assessments of the space, work, and well-being

We then turn to the daily diary findings. Descriptive results of participants' general appraisals of their mission and living space and their daily well-being and life functioning reports are shown in Table 2. In general, participants reported positively for well-being and life and functioning items (i.e., above the midpoint on all 5-point scales). They also rated the space favourably in terms of liveability and functionality.

Interestingly, participants rated the space positively in terms of emotional attachment, as well as for beauty, and loveability, with most participants noting these factors. Despite the generally positive ratings, there was considerable variability across participants and days, with participants using the entire range of the scales, indicating a wide variance in responses, including some negative reports.

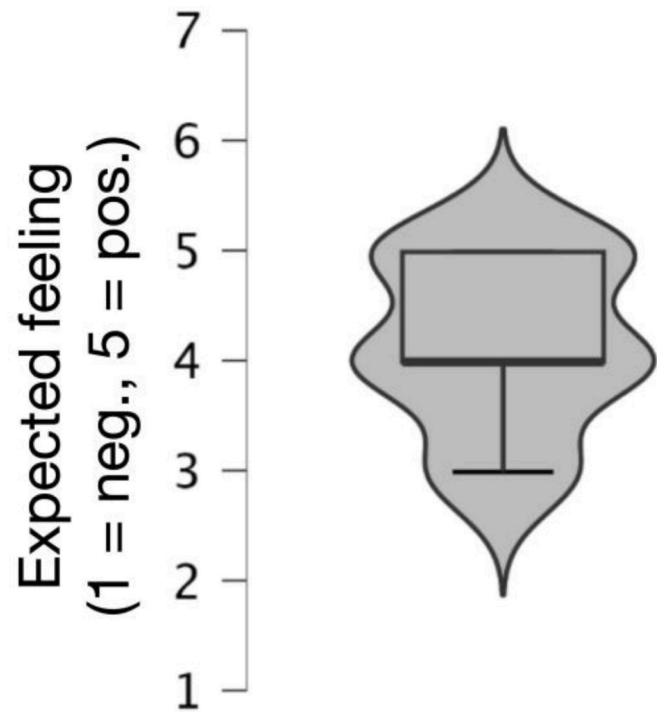


Fig. 6. Participants' expectations regarding their Emotional feeling about Space. (N = 19).

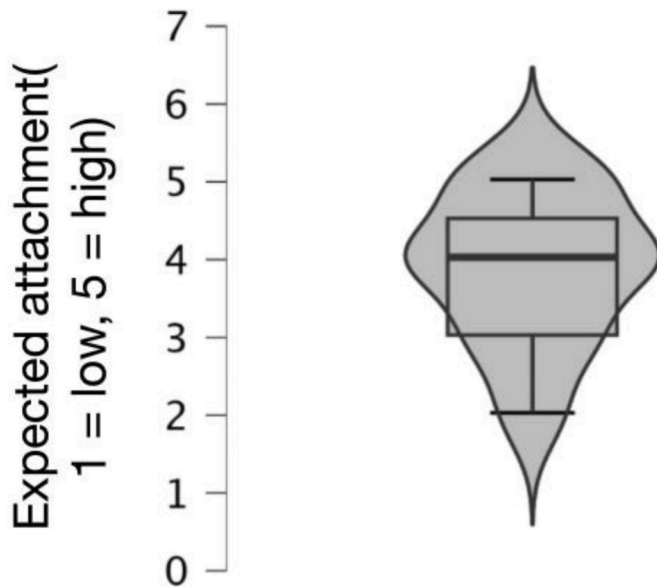


Fig. 7. Participants' expectations regarding their Emotional attachment. (N = 19).

On the other hand, as indicated in Figs. 8–10, which show the mean reports across participants for a selection of factors over the full mission periods, daily reports tended to be rather stable. Notably, we did not find any evidence in any factor for systematic increases or drop-offs, but rather some up-and-down fluctuations.

6.3. Relationship between ratings of space, emotional attachment, and life/work and well-being

We then examined our primary question: the relationship between

Table 2
Descriptive statistics (means and range) of participants' daily ratings of the space, their working and life functioning, and well-being (N = 33 participants, 10–12 days).

daily reported item	M	SD [range]
Well-being/life functioning		
Well-being	3.93	.93 [1–5]
quality of life	3.93	.92 [2–5]
social connectedness	4.01	.99 [1–5]
emotion regulation	4.01	.95 [1–5]
loneliness	1.8	1.03 [1–5]
anxiety	1.86	1.12 [1–5]
work	3.84	.99 [1–5]
efficiency	3.8	.92 [1–5]
social interaction	3.98	.94 [1–5]
relaxation	3.29	1.23 [1–5]
sleep	3.4	1.24 [1–5]
hygiene	3.6	1.14 [1–5]
food	3.82	1.06 [1–5]
leisure	3.12	1.41 [1–5]
space evaluation		
livable	77.78	16.53 [4–100]
functional	79.5	17.69 [7–100]
emotional attachment	67.44	24.87 [2–100]
beautiful	64.48	23.32 [2–100]
lovable	69.3	23.72 [0–100]
interesting	76.4	21.57 [5–100]

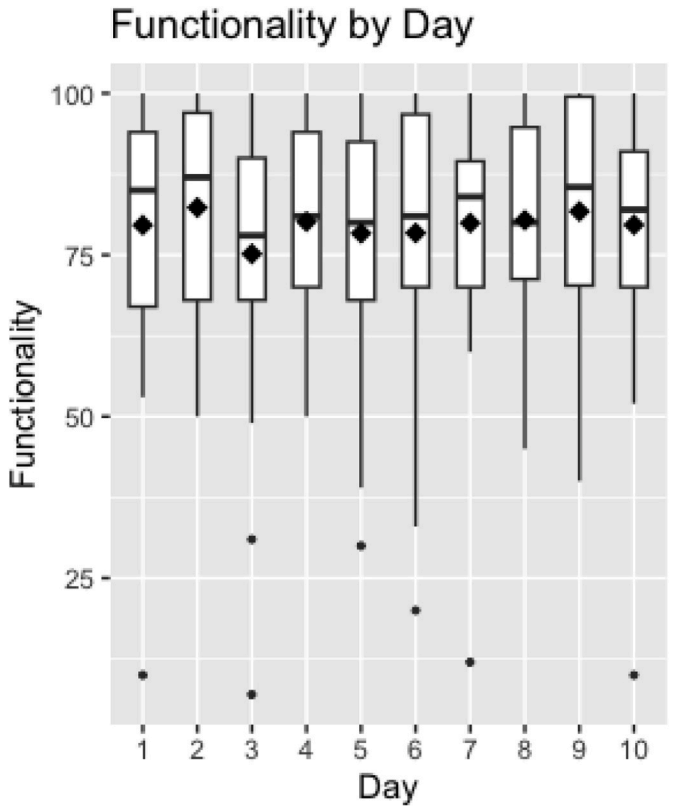


Fig. 8. Repeated measures correlation on functionality.

participants' space ratings and their reported functioning and well-being during the mission. To assess these relationships, we employed a Repeated Measures Correlation. This method (see for further discussion [39]) operates as a standard correlation to assess for related patterns in responses between two factors over a set of participants. However, rather than collapsing across days for each participant for a given rating scale, a repeated measure design allows for the assessment of relations across multiple days while modeling in the natural fluctuations at the

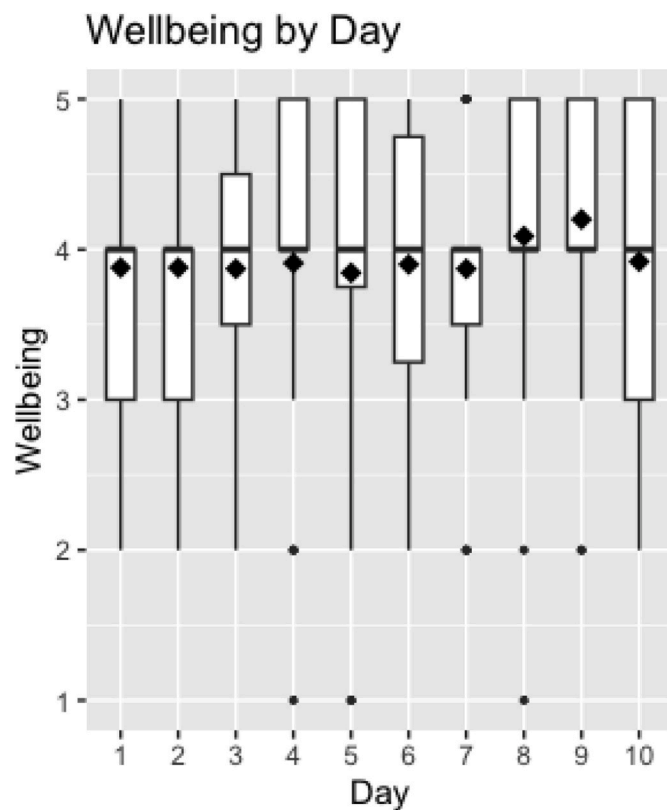


Fig. 9. Repeated measures correlation on well-being by day.

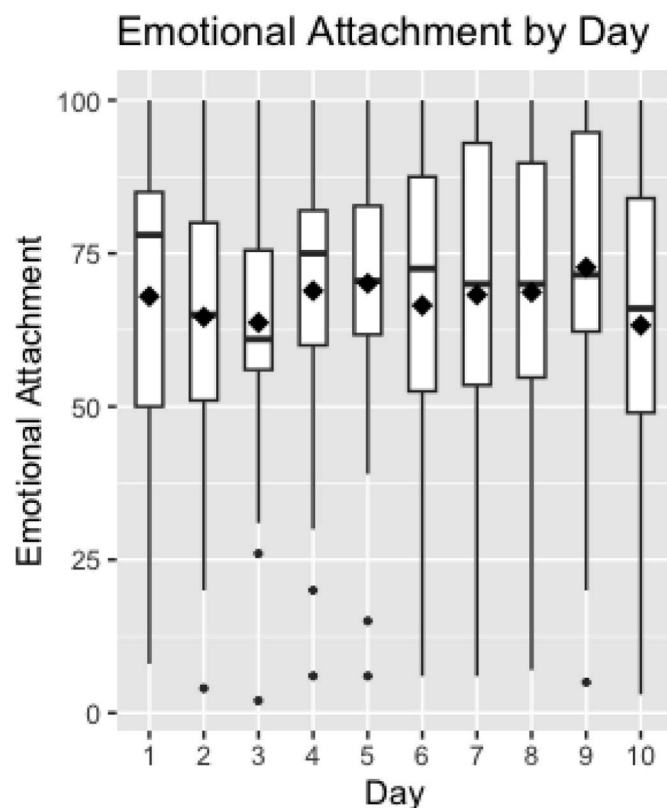


Fig. 10. Repeated measures correlation on emotional attachment by day.

day and participant levels. Assessments were made separately for

specific relations between two given factors across 12 days per participant.

The results, presented in Tables 3 and 4, indicated that ratings of the space as livable and functional were already positively correlated with higher daily well-being, quality of life, and social connections. However, ratings related to higher emotional attachment to the space, as well as perceptions of the space as beautiful and loveable showed even stronger correlations. Notably, emotional attachment and loveability were significantly associated with lower levels of anxiety (for loveable) and loneliness (for both).

Similarly, both liveable/functional and emotional attachment as well as beauty/loveable/interestingness ratings showed significant relations with reported daily efficiency and work, as well as to the quality of relaxation and food. Interestingly, emotional attachment and loveability were more strongly correlated with these factors than livability or functionality.

Both, emotional attachment and loveability, but not livability or functionality, were also linked to better sleep quality. On the other hand, hygiene related mostly to liveability/functionality but not emotional attachment. Although this was related to loveability and beauty.

7. Discussion and conclusion

While some of the findings were expected, others were surprising and provide new insights into the relationship between the built environment, human activities, and psychological well-being. Through our interdisciplinary approach, by combining architectural research with environmental psychology and philosophy, the research confirmed that habitability is more than functionality or operational efficiency. The findings showed that the emotional and aesthetic user experience of the built environment is an integral component of perceived habitability. In addition, the qualitative interviews further contextualised the quantitative findings by expanding on how the crew experienced the habitat and which socio-spatial features they perceived as supporting or constraining their activities and well-being.

One important observation is that, despite the wide range of responses, the variability is significant for our study. It underscores the complexity of human experiences in different environments. For instance, the ratings of work as functional were strongly correlated with participants' perceptions of well-being. This is especially critical when considering anxiety levels; focusing on fostering emotional attachment to the space can be vital in promoting a sense of well-being.

7.1. Work activities - relation to the space

The design of the habitat aims to facilitate efficient work. Participants who perceived the space as functional reported higher levels of well-being. This is not terribly surprising, as the space was designed to support their daily work activities. The comments from the participants emphasize the co-relation of a functional working environment with well-being. "Working on projects [was the most enjoyable] I think. It took so much time, but it was pleasant."

Quite a few participants noted that working on projects and

Table 3

Relations between daily ratings of space and well-being factors (Repeated Measures Correlations), over two week period.

	well-being	QoL	anxiety	loneliness	social connection
livable	.295 ^a	.382 ^a	-.100	-.111	.312 ^a
functional	.144 ^a	.229 ^a	-.086	-.085	.140 ^a
emo. attach	.301 ^a	.374 ^a	-.114	-.165 ^a	.298 ^a
beautiful	.286 ^a	.330 ^a	-.093	-.076	.247 ^a
loveable	.344 ^a	.390 ^a	-.124 ^a	-.170 ^a	.325 ^a
interesting	.193 ^a	.209 ^a	-.043	-.154	.213 ^a

^a Indicates $p < .05$, uncorrected for multiple comparisons due to exploratory nature of the data.

Table 4
Relations between daily ratings of space and life/work function factors (Repeated Measures Correlations), over two week period.

	efficiency	work	relaxation	sleep	hygiene	food
livable	.218 ^a	.295 ^a	.195 ^a	.075	.222 ^a	.230 ^a
functional	.176 ^a	.220 ^a	.133 ^a	.094	.142 ^a	.174 ^a
emo. attach	.304 ^a	.332 ^a	.117 ^a	.131 ^a	.093	.208 ^a
beautiful	.284 ^a	.335 ^a	.204 ^a	.039	.140 ^a	.295 ^a
lovable	.272 ^a	.333 ^a	.174 ^a	.106 ^a	.172 ^a	.212 ^a
interesting	.300 ^a	.330 ^a	.112 ^a	.071	.075	.234 ^a

^a Indicates $p < .05$, uncorrected for multiple comparisons due to exploratory nature of the data.

conducting EVAs (Extravehicular Activities) were among the most enjoyable aspects of their experience (Fig. 11). One participant remarked: “What I liked the most was the projects and EVA; it was great, and I worked until late at night to finish them. They were interesting and even demanding.” Others noted: “EVA and project parts were more enjoyable because I learned a lot.” And “I really enjoyed the EVAs.”

These reflections highlight the positive connection between the workspace and participants ability to relate to their work in a meaningful manner. The personal interviews highlighted the importance of the perceived quality of the working environment in terms of its support for meaningful work and professional identity. One participant described how she spent most of the time in the engineering workshop and operations room because these spaces facilitated working together with fellow crew members and also to focus on the specific work tasks. She noted that these team work sessions were more relevant than other mission activities, arguing that architectural spaces gain meaning through the human activities and social interactions they enable.

7.2. Food activities – relation to the space

The relevance of food, livability, and functionality within the habitat is also not surprising as it is also known from historical anecdotes from astronauts (cf. [40]).

During most missions some crewmembers prepared lunch and dinner for the entire crew, while breakfast was typically prepared individually. As one participant noted, “Having a crew member who acted as the chef was certainly a big plus.” Fig. 12 shows two crewmembers preparing food in the kitchen and dining room. The image shows that food preparation with communal meals are essential aspects of daily life in an isolated environment. One participant reflected on the significance of these moments: “The best part of the mission for me was the day off. Then, together with the crew, we prepared Polish food. For me, it was the most



Fig. 11. EVA Activities on the simulated lunar surface. Credits: LunAres.

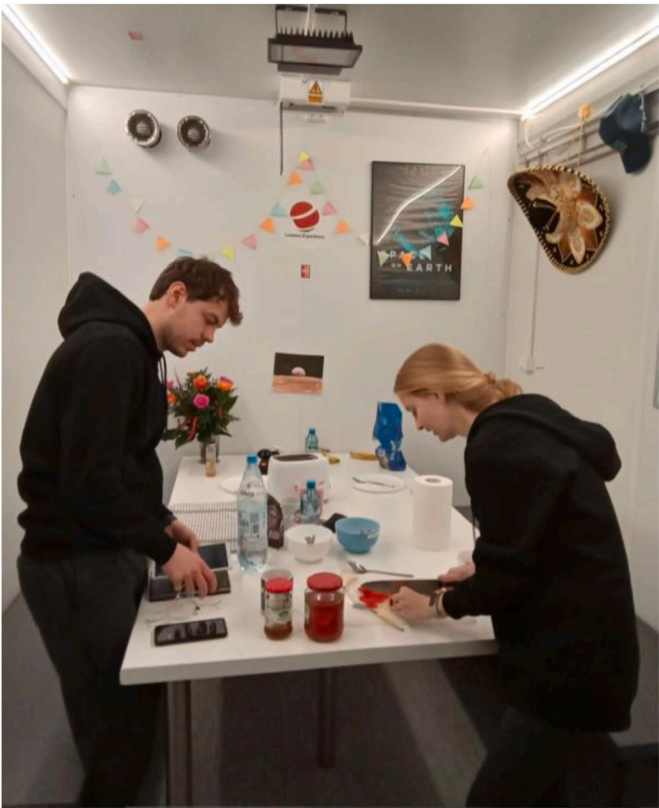


Fig. 12. The kitchen and dining room. Credit: Sara Borjanovic.

pleasant moment because we spent this time together preparing a meal, talking, and laughing with each other. All this happened in the kitchen, so I felt right at home. The most important things always happen in the kitchen.” (Study Participant, Study S.P.A.C.E.S). This citation illustrates how food-related activities become more than simple nutrition activities. They can also foster emotional attachment to the habitat because of shared experiences and social interaction. This also adds a sense of belonging. The interviews further revealed that operational constraints, such as limited water availability and hygiene routines, became an integral part of everyday life and influenced how participants experienced the activities taking place in the habitat. They especially mentioned food preparation and communal living. Rather than being perceived solely as restrictions - because they did not come as a surprise, but were expected - these routines often reinforced teamwork and social cohesion. This is another indicator that demonstrates how environmental aspects of the habitat and their emotional experience became intertwined.

7.3. Leisure activities - relation to the space

Participants expressed some dissatisfaction with leisure possibilities during their mission. The physical limitations of the habitat has most probably contributed to a sense of confinement, impacting their ability to fully relax. Fig. 13 shows crew members exercising together as a joint group activity. This area requires further exploration to better understand how habitat design can better support and enhance leisure activities.

The interviews also underscored that informal social leisure activities played an important role in maintaining crew morale. Participants shared their experiences of community celebrations, movie evenings, storytelling sessions, and other informal gatherings as valuable opportunities to strengthen interpersonal relationships. They have also seen these activities as a possibility to temporarily shift attention away from operational and routine tasks. Further, these activities helped transform the habitat from a purely functional workplace into a shared living

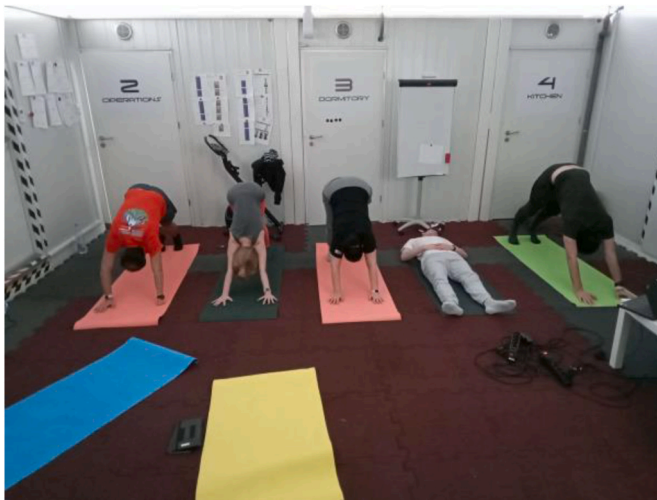


Fig. 13. Doing yoga in the atrium to stay fit, relax, stretch and recover faster from the additional daily MaxForce resistance training.

space.

7.4. Sleep activities - relation to the space

The duration of sleep was notably short during the missions, and that could have influenced general well-being. While loneliness did not emerge as a significant predictor, the sense of social connectedness was crucial. Interestingly, while social interaction did not significantly impact well-being, the overall duration of missions may affect participants' feelings of connection and contentment. Fig. 14 shows the personal arrangement of one of the crewmembers for their sleep station.

Beyond the sleeping quarters themselves, the interviews highlighted the importance of providing opportunities for privacy and temporary retreat. One participant suggested that future habitats should incorporate small semi-private spaces where crew members could hold confidential conversations or simply withdraw temporarily from the group. Such spaces may become increasingly important during long-duration missions, where opportunities for psychological recovery and conflict management are limited. These results indicate that privacy is not merely a matter of sleeping accommodation, but a more general quality of architecture contributing to the individual well-being while permitting social cohesion in confined habitats.

7.5. Emotional attachment to the built environment

Perhaps the most important contribution of this study is the finding that emotional attachment to the built environment explained aspects of crew experience that could not be fully accounted for by traditional habitability measures such as functionality or liveability alone. Emotional attachment to the space emerged as a significant factor influencing participants' experiences. The qualitative feedback reflects this connection, as many participants expressed how their attachment to the environment impacted their overall well-being and work performance.

Even more, this aspect showed often higher relations to both working and well-being aspects, as well as relations to other factors where consideration of liveability and functionality were not found - for example in aspects such as anxiety and loneliness.

These findings expand the authors' previous work on habitability and human-centered habitat design, which has mostly focused on human activities, and socio-spatial design principles (cf. [2,3]). They further strengthen the growing recognition that the psychosocial qualities of the built environment should be considered alongside functional requirements when designing future space habitats (cf. [41]).

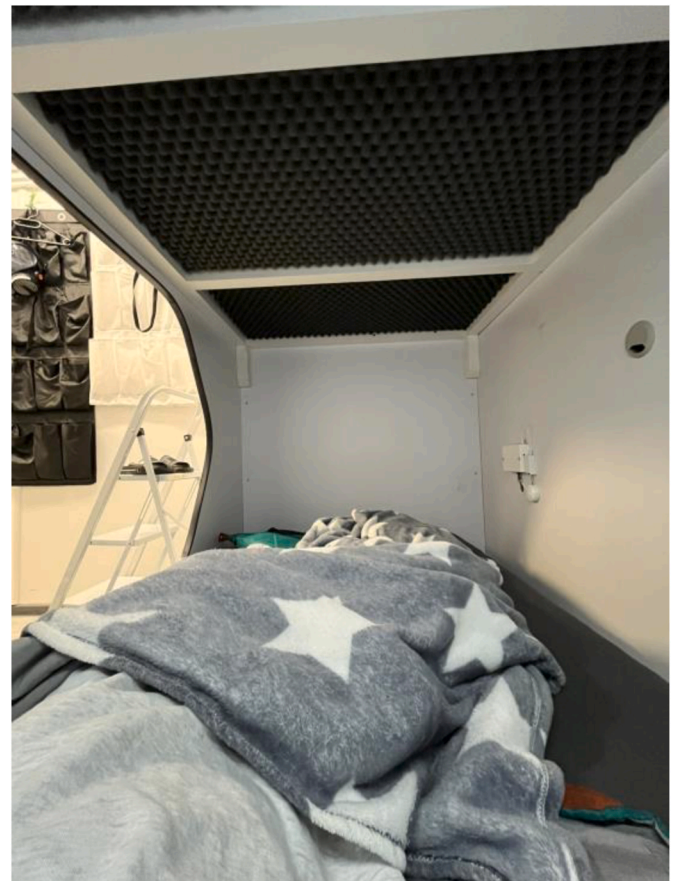


Fig. 14. Inside of a bottom sleeping pod - limited space with only one shelf above head, light and some acoustic absorbers on top to reduce the snoring noises during night.

This study and collaboration in research could also have relevant implications for terrestrial architecture, especially as urban density increases and demands arise for built environments to not only provide operational efficiency but also psychological well-being and emotional connection (cf. [42]).

CRediT authorship contribution statement

Sandra Haeuplik-Meusburger: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Matthew Pelowski:** Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Joerg Fingerhut:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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